Lecture 24 — Inverses from elementary matrices

Mon Nov 2

Last updated: October 29, 2026.

Boyd video (optional): Lecture 30 — left & right inverses · ENGR108 playlist (Boyd)

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Reading: Boyd 11.1–11.3; recall elementary matrices from Lecture 21. Related objectives: Lectures. Schedule. Lab 3 already peeked at $[A \mid I]$ for the nutrient matrix — today we name the right block A^{-1} and make the idea general.

Bring paper. Warm-up, fill-in notes, board examples, and **Your turn**.

Learning objectives

By the end of class, you should be able to:

1. Explain A^{-1} as the product of elementary matrices that reduce A to I.
2. Derive and use the 2×2 inverse formula from elimination.
3. Compute a 3×3 inverse by eliminating $[A \mid I]$ to $[I \mid A^{-1}]$ (when it exists).
4. Check a candidate inverse with $A^{-1}A = I$ (or $AA^{-1} = I$).
5. Explain why A^{-1} is a *general* undo: the same matrix gives $x = A^{-1}b$ for every b.
6. Recognize when no inverse exists, and connect invertibility to unique solvability of $Ax = b$.

Warm-up

In Lab 3 you reduced $[A \mid I]$ for the nutrient matrix and got a right block B with $BA = I$ and $Bb = x$ for every meal target b you tried.

(a) What name should B get?

(b) In Lecture 21 we had $E_3E_2E_1A = I$. How does that product relate to B?

Fill in

Inverse from elementary matrices

A square matrix A is **invertible** if there exist elementary matrices $E_1, \ldots, E_k$ (row operations) with

$$E_k \cdots E_2E_1A = _____.$$

We then define

$$A^{-1} := E_k \cdots E_2E_1.$$

So $A^{-1}A = I$. (When A is invertible one also gets $AA^{-1} = I$.)

Meaning. Each E_i undoes / performs one reversible row operation. Their product undoes the whole map $x \mapsto Ax$: if $y = Ax$, then $A^{-1}y = x$.

Solving — why the inverse is powerful. If A is invertible and $Ax = b$, then

$$x = _____.$$

The point of A^{-1} is **generality**: you build one matrix that works for **any** right-hand side. Change b, keep the same A^{-1}, multiply again. Equivalently: $Ax = b$ has a **unique** solution for every b iff A is invertible.

(Elimination on $[A \mid b]$ still finds one x for one b. The inverse is the packaged "undo everything" map.)

Same E_i, two packages (Lab 3). Stopping elimination at an upper-triangular U gives a factorization $A = LU$ — great for **solving** many systems with the same A. Pushing all the way to I on $[A \mid I]$ builds A^{-1} — great for the **undo map**. Software's `A\b` is closer to the LU idea than to forming A^{-1} first.

Practical computation

To find A^{-1} by elimination, form the wide matrix $[A \mid I]$ and row-reduce until the left block is I (when possible). The right block is then _____.

For 2×2, that procedure yields a closed formula (next). For 3×3, do the elimination by hand on $[A \mid I]$.

Check. Always verify $A^{-1}A = I$ (or $AA^{-1} = I$) on a candidate.

When it fails. If a zero row appears on the left while reducing $[A \mid I]$, you cannot reach I: A has _____ inverse. (Columns are dependent; $Ax = 0$ has a nontrivial solution.)

Worked in class

W1. Product of E_i is the inverse. Let

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}.$$

(a) Write the elementary matrix E that subtracts 2 times row 2 from row 1 (after you already have a pivot 1 in position $(2, 2)$ — or clear the $(1, 2)$ entry of A directly). Check $EA = I$.

(b) Conclude $A^{-1} = E$ and write A^{-1} explicitly.

W2. Derive the 2×2 formula. Let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

with $ad - bc \neq 0$. Reduce $[A \mid I]$ to $[I \mid A^{-1}]$ (or clear systematically with elementary matrices) and show

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

(We will call $ad - bc$ the **determinant** on Friday.)

W3. Use the formula. Compute A^{-1} for

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$$

using W2. Check $A^{-1}A = I$.

W4. Another 2×2. Find A^{-1} for

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

by the formula, then check $AA^{-1} = I$.

W5. 3×3 by elimination. Form $[A \mid I]$ for

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

and reduce the left block to I. Read A^{-1} from the right block. Check $A^{-1}A = I$.

W6. No inverse. Try the same method for

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}.$$

What goes wrong in $[A \mid I]$? Connect to $Ax = 0$ or dependent columns.

W7. When does an inverse exist? For square A, list equivalent ideas (any three):

- can reduce A to I by elementary matrices
- A^{-1} exists
- $Ax = b$ has a unique solution for every b
- $Ax = 0$ has only the zero solution
- columns of A are independent
- (Friday) $\det A \neq 0$

W8. Same inverse, two right-hand sides. For

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix},$$

form A^{-1} (formula). Then compute $x = A^{-1}b$ for

$$b = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \quad \text{and} \quad b' = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Check $Ax = b$ and $Ax' = b'$. Same A^{-1}, two different answers — that is the "general undo" idea.

Your turn

Y1. Using elementary matrices / $[A \mid I]$, find A^{-1} for

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Check $A^{-1}A = I$.

Y2. Use the 2×2 formula on

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

(90° CCW). What geometric map is A^{-1}?

Y3. Reduce $[A \mid I]$ for

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

and read A^{-1}. Check one of $A^{-1}A$ or AA^{-1}.

Y4. Does

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

have an inverse? Explain using $[A \mid I]$ or $Ax = b$.

Y5. True or false, with a one-line reason: if A is invertible, then for every b the system $Ax = b$ has exactly one solution.

Y6. True or false, with a one-line reason: if you already know A^{-1}, you can solve $Ax = b$ for a *new* b without redoing elimination on A — just multiply $A^{-1}b$.

Answers (Your turn)

Y1. Clear the $(1,2)$ entry: $E = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$ gives $EA = I$, so

$$A^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}.$$

Check: $A^{-1}A = I$.

Y2. $ad - bc = 1$, so

$$A^{-1} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

(rotation 90° clockwise). Check: $A^{-1}A = I$.

Y3. $[A \mid I]$ reduces to right block

$$A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(subtract 2 times row 1 from row 2 on the wide matrix). Check: $A^{-1}A = I$.

Y4. **No.** Left block of $[A \mid I]$ gets a zero row; cannot reach I. Equivalently $Ax = (0,1)$ has no solution, or $Ax = 0$ has nontrivial solutions such as $(0,1)$.

Y5. **True.** Invertibility means $x = A^{-1}b$ is the unique solution for every b.

Y6. **True.** That is the point of A^{-1}: one general undo for every b.